

Simple AWS Auto Scaling Project

Phase 1

Create EC2 instance

- Choose Amazon predefined AMI
- Choose instance type
- Create or choose Key pair

EC2 > Instances > Launch an instance

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0685bcc683dad6b69 / ami-067770fe0500a2d57 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260526.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	uefi-preferred	ami-0685bcc683dad6b69	2026-05-21	ec2-user

Verified provider

▼ Instance type Info | Get advice

Instance type

t2.nano
Family: t2 1 vCPU 0.5 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 0.008 USD per Hour On-Demand SUSE base pricing: 0.0062 USD per Hour
On-Demand Linux base pricing: 0.0062 USD per Hour On-Demand Windows base pricing: 0.0085 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Summary

Number of instances | Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11.....read more
ami-0685bcc683dad6b69

Virtual server type (instance type)

t2.nano

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Network ,Storage and Firewall Settings

- Select VPC
- Enable Public IP
- Enable SSH (22) and HTTP (80) ports in Security group
- Choose root volume or additional storage (if required file system)

EC2 > Instances > Launch an instance

VPC - required Info

vpc-0d9bf3ad2f4293e02 (Mumbai_VPC-vpc)
10.0.0.0/16

Subnet Info

subnet-0e4b21eeea44fd6c9 Mumbai_VPC-subnet-public1-ap-south-1a
VPC: vpc-0d9bf3ad2f4293e02 Owner: 639713291467 Availability Zone: ap-south-1a (aps1-a21)
Zone type: Availability Zone IP addresses available: 4091 CIDR: 10.0.0.0/20

Create new subnet

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

launch-wizard-2

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and _-./#,@!+*%&(){}\$*

Description - required Info

launch-wizard-2 created 2026-06-01T15:14:43.562Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0, SSH for instance access) Remove

Type	Protocol	Port range
ssh	TCP	22

Source type Info

Anywhere

Source Info

Q Add CIDR, prefix list or security group
0.0.0.0/0 X

Description - optional Info

SSH for instance access

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0, HTTP for web access) Remove

Type	Protocol	Port range
HTTP	TCP	80

Source type Info

Anywhere

Source Info

Q Add CIDR, prefix list or security group
0.0.0.0/0 X

Description - optional Info

HTTP for web access

▼ Summary

Number of instances | Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.11.....read more
ami-0685bcc683dad6b69

Virtual server type (instance type)

t2.nano

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

▼

Configure storage

Info ⓘ

Advanced

1x

8

GIB

gp3

▼

Root volume, 3000 IOPS, Not encrypted

Add new volume

🕒

Click refresh to view backup information

↻

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

☐

S3 Files - new

☐

FSx

☒

None

☐

EFS

▶

Advanced details

Info ⓘ

- Insert User data (initial script)

User data - optional

Info ⓘ

Upload a file with your user data or enter it in the field.

⬆

Choose file

```
#!/bin/bash

sudo dnf update -y

sudo dnf install nginx -y
sudo systemctl enable nginx
sudo systemctl start nginx

sudo mkdir /data
```

- Create additional volume as per requirement

EC2 > Volumes > Create volume

Create volume

Info ⓘ

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type

Info ⓘ

General Purpose SSD (gp3)

▼

Size (GiB)

Info ⓘ

10

Min: 1 GiB, Max: 65536 GiB.

IOPS

Info ⓘ

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s)

Info ⓘ

125

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Availability Zone

Info ⓘ

aps1-az1 (ap-south-1a)

▼

Snapshot ID - optional

Info ⓘ

Don't create volume from a snapshot

▼

🕒

Encryption

Info ⓘ

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

Tags - optional

Info ⓘ

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

- Attach volume to EC2 instance

EC2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

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Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

Successfully created volume vol-06c3e64299a0d48f2.

Volumes (1/2)

External volume

vol-06c3e64299a0d48f2

gp3

10 GiB

3000

125

-

-

vol-08bef32a4ef639a5e

gp3

8 GiB

3000

125

snap-06830fb...

-

Attach volume

Modify volume

Create snapshot

Create snapshot lifecycle policy

Delete volume

Detach volume

Force detach volume

Manage auto-enabled I/O

Copy volume - new

Manage tags

Resilience testing

Previously fault injection

Volume ID: vol-06c3e64299a0d48f2 (External volume)

Details

Status checks

Monitoring

Tags

Volume ID

vol-06c3e64299a0d48f2 (External volume)

Size

10 GiB

Type

gp3

Status check

OKay

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Volume state

Available

IOPS

3000

Throughput

125

Fast snapshot restored

No

Availability Zone

aps1-az1 (ap-south-1a)

Created

Mon Jun 01 2026 21:23:49 GMT+0530 (India Standard Time)

Multi-Attach enabled

No

Attached resources

-

Outposts ARN

-

Managed

false

Operator

-

Source

Snapshot ID

Source volume ID

lumes > vol-06c3e64299a0d48f2 > Attach volume

Attach volume

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

vol-06c3e64299a0d48f2 (External volume)

Availability Zone

aps1-az1 (ap-south-1a)

Instance

I-019d1d7991fc94eec

(My web server) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name

/dev/sdf

Recommended device names for Linux: /dev/xvda for root volume, /dev/sd(f-p) for data volumes.

Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

Cancel

Attach volume

- Verify EC2 availability in the AWS console.

Instances (1/1)

My web server

I-019d1d7991fc94eec

Running

t2.nano

2/2 checks passed

ap-south-1a

ec2-65-2-34-124.ap-so...

65.2.34.124

-

-

i-019d1d7991fc94eec (My web server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

I-019d1d7991fc94eec

IPV6 address

-

Hostname type

IP name: ip-10-0-8-180.ap-south-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

65.2.34.124 [Public IP]

IAM role

EC2-WebServer-Role

IMDSv2

Required

Operator

-

Public IPv4 address

65.2.34.124 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-8-180.ap-south-1.compute.internal

Instance type

t2.nano

VPC ID

vpc-0d9bf3ad2f4293e02 (Mumbai_VPC-vpc)

Subnet ID

subnet-0e4b21eeea44fd6c9 (Mumbai_VPC-subnet-public1-ap-south-1a)

Instance ARN

arn:aws:ec2:ap-south-1:639713291467:instance/i-019d1d7991fc94eec

Private IPv4 addresses

10.0.8.180

Public DNS

ec2-65-2-34-124.ap-south-1.compute.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Access EC2 instance

- Check storage
- Check Nginx service status

[illegible]

- Create file system and mount external storage using below commands
 - lsblk : show available storage
 - mkfs.ext4 /dev/xvdf : create file system of additional storage
 - mount /dev/xvdf /data : Mount created file system in over local folder
 - Sudo vi /etc/fstab : Edit file system file and add external storage UUID, type, etc..
 - UUID="" /mount_location ext4 defaults,nofail 0 2
 - Sudo mount -a

```
[ec2-user@ip-10-0-8-180 ~]$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0     8G  0 disk
├─xvda1     202:1    0     8G  0 part /
├─xvda127   259:0    0     1M  0 part
└─xvda128   259:1    0    10M  0 part /boot/efi
xvdf        202:80   0    16G  0 disk
zram0       253:0    0   458M  0 disk [SWAP]
[ec2-user@ip-10-0-8-180 ~]$ sudo mkfs.ext4 /dev/xvdf
mkfs.ext4 1.46.5 (30-Dec-2021)
Creating filesystem with 2621440 4k blocks and 655360 inodes
Filesystem UUID: 5298fd9f-01b2-40f0-a712-f298b6c58eb2
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-10-0-8-180 ~]$ ls /
bin boot data dev etc home lib lib64 local media mnt opt proc root run sbin srv sys usr var
[ec2-user@ip-10-0-8-180 ~]$ sudo mount /dev/xvdf /data
[ec2-user@ip-10-0-8-180 ~]$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0     8G  0 disk
├─xvda1     202:1    0     8G  0 part /
├─xvda127   259:0    0     1M  0 part
└─xvda128   259:1    0    10M  0 part /boot/efi
xvdf        202:80   0    16G  0 disk /data
zram0       253:0    0   458M  0 disk [SWAP]
[ec2-user@ip-10-0-8-180 ~]$
```

Phase 3

- Create website folder in external mounted storage

```
[ec2-user@ip-10-0-8-180 www]$ pwd
/data/ntlm/www
[ec2-user@ip-10-0-8-180 www]$
```

- Create index.html file inside this folder

```
background-color: #e68a00;
cursor: pointer;
}

footer {
background-color: var(--primary-color);
color: #0a0e0d;
text-align: center;
padding: 1.5rem;
font-size: 0.9rem;
margin-top: auto;
}
</style>
</head>
<body>
<header>
<h1>AWS Cloud Practice Deployment</h1>
<p>Testing High Availability Website with Load Balancer with Auto Scaling Group Infrastructure</p>
<div class="badge">HTTP Active</div>
</header>
<main>
<section class="card">
<h2>Success! Your Web Server is Live</h2>
<p>If you can see this page over a secure URL (starting with <strong>https://</strong>), it means your SSL/TLS certificates, security groups, and web server configurations are working properly.</p>
<ul class="status-list">
<li><strong>EC2 Instance:</strong> Create Web server.</li>
<li><strong>IAM Role:</strong> Use for limited access</li>
<li><strong>Security Groups:</strong> Configured to allow HTTP (80)</li>
<li><strong>ELB:</strong> Use external storage for store website data (prevent data loss).</li>
<li><strong>ALB:</strong> Application Load Balancer</li>
<li><strong>Web Server:</strong> Document root successfully updated.</li>
</ul>
<a href="#" class="btn" onclick="alert('Verification successful! Keep up the great work learning AWS.')">Verify Connection</a>
</section>
</main>
<footer>
<p><strong>copy</strong>; 2026 AWS Practice Lab | Built with HTML & CSS</p>
</footer>
</body>
</html>
```

i-019d1d7991fc94eec (My web server)
PublicIPs: 65.2.34.124 PrivateIPs: 10.0.8.180

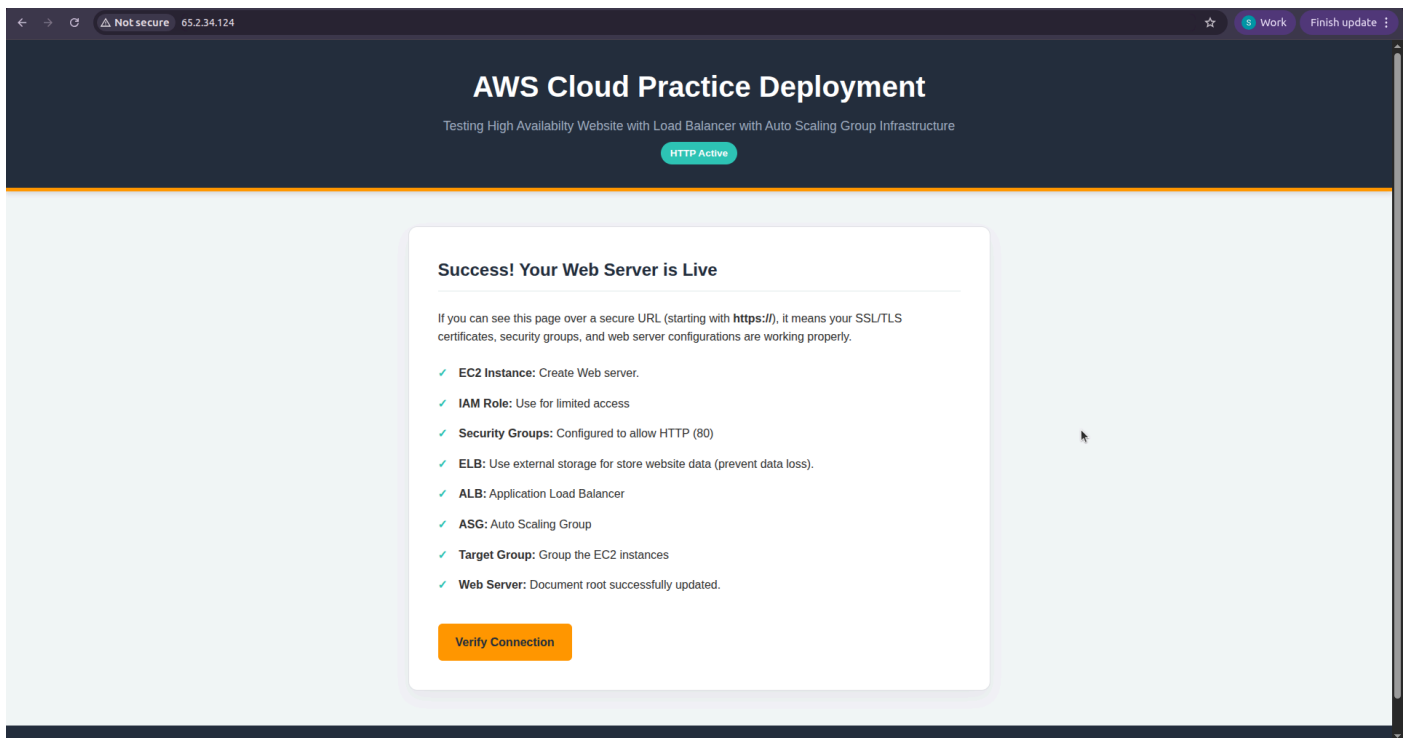
- Edit nginx config file
 - Edit root folder location: replace default location to our website folder

```
server {
listen 80;
listen [::]:80;
server_name _;
root /data/html/www;

# Load configuration files for the default server block.
include /etc/nginx/default.conf;
}
```

- Restart nginx service after configuration changes

- Website is live



Phase 2

Create Amazon Machine Image (AMI - Golden Image)

- Select EC2 instance to create AMI
- Go to Action -> Image and template -> create image

The screenshot shows the AWS Management Console interface. At the top, the 'Instances (1/1)' page is visible, showing a single instance named 'My web server' with ID 'i-019d1d7991fc94eec' in a 'Running' state. The 'Actions' menu is open, and 'Create image' is selected. Below this, the 'Create image' wizard is shown. The 'Image details' section includes fields for 'Instance ID' (i-019d1d7991fc94eec), 'Image name' (Web-server-AMI), and 'Image description' (AMI include NGINX server with hosted website). The 'Reboot instance' checkbox is checked. Under 'Instance volumes', two volumes are listed: a root volume of type 'EBS' on device '/dev/xv...' with a size of 8 GB, and a data volume of type 'EBS' on device '/dev/sdf' with a size of 10 GB. Both volumes are set to 'Delete on termination' and 'Encrypted'. A note states: 'During the image creation process, Amazon EC2 creates a snapshot of each of the above volumes.' The bottom of the console shows the footer with '© 2026, Amazon Web Services, Inc. or its affiliates.'

- Verify AMI availability in the AWS console.

The screenshot shows the 'Amazon Machine Images (AMIs) (1/1)' page in the AWS Management Console. A table lists the AMIs, with one entry: 'WEB-AMI' with name 'Web-server-AMI', ID 'ami-0127066faa399ce15', source '639713291467/Web-server-AMI', owner '639713291467', visibility 'Private', status 'Pending', and creation date '2026/06/01 21:54 GMT+5:30'. Below the table, the 'Details' tab for the selected AMI is shown. It includes fields for 'AMI ID' (ami-0127066faa399ce15), 'AMI name' (Web-server-AMI), 'Image type' (machine), 'Owner account ID' (639713291467), 'Platform details' (Linux/UNIX), 'Architecture' (x86_64), 'Root device type' (EBS), and 'Usage operation' (RunInstances).

Phase 3

Create EC2 Launch Template

- Choose AMI from phase 2
- Choose instance type as per requirement

EC2 > Launch templates > Create launch template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ Application and OS Images (Amazon Machine Image) info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

☐ Don't include in launch template

☒ Owned by me

☐ Shared with me

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Web-server-AMI

ami-0127066faa399ce15

2025-06-01T16:24:10.000Z

Virtualization: hvm

ENA enabled: true

Root device type: ebs

Boot mode: uefi-preferred

Description

AMI Include NGINX server with hosted website

Architecture

AMI ID

x86_64

ami-0127066faa399ce15

▼ Instance type info | Get advice

Advanced

Instance type

t2.nano

Family: t2

1 vCPU

0.5 GiB Memory

Current generation: true

On-Demand Ubuntu Pro base pricing: 0.008 USD per Hour

On-Demand SUSE base pricing: 0.0062 USD per Hour

On-Demand Linux base pricing: 0.0062 USD per Hour

On-Demand Windows base pricing: 0.0085 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Summary

Software image (AMI)

AMI Include NGINX server with ...[read more](#)

ami-0127066faa399ce15

Virtual server type (instance type)

t2.nano

Firewall (security group)

launch-wizard-1

Storage (volumes)

2 volume(s) - 18 GiB

Cancel

Create launch template

- Choose Network setting and Enable auto assign public IP option from advanced network configuration

EC2 > Launch templates > Create launch template

▼ Network settings info

Subnet info

subnet-0e4b21eeea44fd6c9

Mumbai_VPC-subnet-public1-ap-south-1a

[Create new subnet](#)

VPC: vpc-0d9b0f3ad2f4293e02

Owner: 639713291467

Availability Zone: ap-south-1a (aps1-az1)

Zone type: Availability Zone

IP addresses available: 4090

CIDR: 10.0.0.0/20

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone info

Don't include in launch template

[Enable additional zones](#)

Firewall (security groups) info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group

☐ Create security group

Common security groups info

Select security groups

launch-wizard-1 sg-02946fe197f932477

[X](#)

[Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Summary

Software image (AMI)

AMI Include NGINX server with ...[read more](#)

ami-0127066faa399ce15

Virtual server type (instance type)

t2.nano

Firewall (security group)

launch-wizard-1

Storage (volumes)

2 volume(s) - 18 GiB

Cancel

Create launch template

- Verify Launch template in AWS console

Launch Templates (1/1) [Info](#)

Search

Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By	Managed	Operator
lt-0ab2b1e1b6dbe51fb	Web-server-launch-template	1	1	2026-06-01T16:40:44.000Z	arn:aws:iam::639713291467:root	false	-

Web-server-launch-template (lt-0ab2b1e1b6dbe51fb)

Launch template details

Launch template ID: lt-0ab2b1e1b6dbe51fb | Launch template name: Web-server-launch-template | Default version: 1 | Owner: arn:aws:iam::639713291467:root

[Actions](#) [Delete template](#)

[Details](#) | [Versions](#) | [Template tags](#)

Launch template version details

Version: 1 (Default) | Description: V1 | Date created: 2026-06-01T16:40:44.000Z | Created by: arn:aws:iam::639713291467:root

[Actions](#) [Delete template version](#)

[Instance details](#) | [Storage](#) | [Resource tags](#) | [Network interfaces](#) | [Advanced details](#)

AMI ID: ami-0127066faa399ce15 | Instance type: t2.nano | Availability Zone: - | Availability Zone id: -

Phase 4

Create target group

- Choose target type as per our requirement
- Choose required protocol
- Choose IP address type
- Select VPC

[EC2](#) > [Target groups](#) > Create target group

Step 2 - recommended
☐ Register targets
 Step 3
☐ Review and create

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable
 Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.*

Target type
 Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
 Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or EC2 instances for automatic management.
 Suitable for: ALB, NLB, GWLB

☐ **IP addresses**
 Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
 Suitable for: ALB, NLB, GWLB

☐ **Lambda function**
 Supports load balancing to a single Lambda function. ALB required as traffic source.
 Suitable for: ALB

☐ **Application Load Balancer**
 Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
 Suitable for: NLB

Target group name
 Name must be unique per Region per AWS account.
 Web-server-target-group
 Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 23/32

Protocol
 Protocol for communication between the load balancer and targets.
 HTTP

Port
 Port number where targets receive traffic. Can be overridden for individual targets during registration.
 80
 1-65535

IP address type
 Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
 Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
 Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
 Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.
 vpc-0d9b3ad2f4293e02 (Mumbai_VPC-vpc)
 10.0.0.0/16

[Create VPC](#)

- Choose Health check Protocol
- Give the health check path (default root[/])
- Edit Advanced check as per requirement

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP

Health check path

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/

Up to 1024 characters allowed.

► Advanced health check settings

• Add EC2 instance in target group

Register targets - *recommended*

This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (1)

Filter instances

☐	Instance ID	Name	State	Security groups	Zone	Private IPv4 address	Subnet
☐	i-019d1d7991fc94eec	My web server	Running	launch-wizard-2	ap-south-1a	10.0.8.180	subnet-

0 selected

Ports for the selected instances

Ports for routing traffic to the selected instances.

80

1-65535 (separate multiple ports with commas)

Include as pending below

1 selection is now pending below. Include more or register targets when ready.

Review targets

Targets (1)

Filter targets

Show only pending

Remove all pending

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address	Subnet ID	Launch time
i-019d1d7991fc94eec	My web server	80	Running	launch-wizard-2	ap-south-1a	10.0.8.180	subnet-0e4b21eeea44fd6c9	June 1, 2026, 21:22 (UTC+05:30)

• Target group review

EC2 > Target groups > Create target group

- Step 1
Create target group
- Step 2 - recommended
Register targets
- Step 3
Review and create

Review and create

Review your target group configuration before creating

Step 1: Target group details

Edit

Target group details

Name
Web-server-target-group

VPC
[vpc-0d9bf3ad2f4295e02](#)

Target type
Instance

IP address type
IPv4

Protocol : Port
HTTP: 80

Protocol version
HTTP1

Health check details

Health check protocol
HTTP

Timeout
5 seconds

Health check path
/

Healthy threshold
5

Health check port
traffic-port

Unhealthy threshold
2

Interval
30 seconds

Success codes
200

Step 2: Register targets

Edit

Targets (1)

Instance ID	Name	Port	Zone
i-019d1d7991fc94eec	My web server	80	ap-south-1a

Cancel

Previous

Create target group

- Verify target group in AWS Console

Web-server-target-group

Successfully created the target group: Web-server-target-group. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.

Web-server-target-group

Details

arn:aws:elasticloadbalancing:ap-south-1:639713291467:targetgroup/Web-server-target-group/a9b9df80cbdf1894

Target type

Instance

Protocol : Port

HTTP: 80

Protocol version

HTTP1

VPC

vpc-0d9bf3ad2f4293e02

IP address type

IPv4

Load balancer

None associated

1

Total targets

0

Healthy

0

Unhealthy

1

Unused

0

Initial

0

Draining

0

Anomalous

Distribution of targets by Availability Zone (AZ)

Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets

Monitoring

Health checks

Attributes

Tags

Registered targets (1)

Anomaly mitigation: Not applicable

Deregister

Register targets

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

1

	Instance ID	Name	Port	Zone	Health status	Health status details	Administrative o...	Override details	Launch...	Anomaly
☐	i-019d1d7991fc94eec	My web server	80	ap-south-1a (a...	Unused	Target group is not co...	-	-	June 1, 20...	Normal

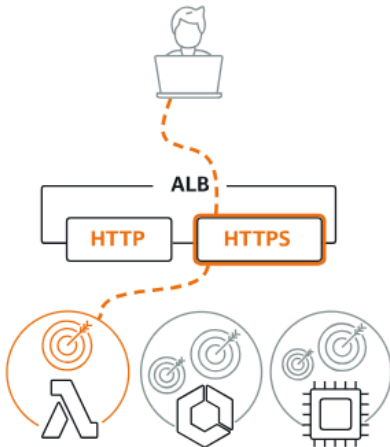
Phase 5

Create Load Balancer

- Choose type of load balance (ALB)
- Choose scheme (internet or intranet facing)

Load balancer types

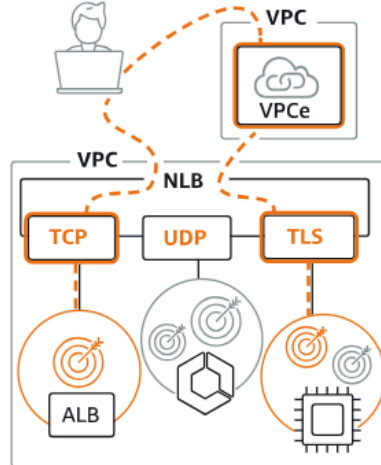
Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

Create

Network Load Balancer [Info](#)



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

Create

Gateway Load Balancer [Info](#)



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

Create

► Classic Load Balancer - *previous generation*

- ALB Network mapping and choose AZs

EC2 > Load balancers > Create Application Load Balancer

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0d9bf5ad2f4293e02 (Mumbai_VPC-vpc)
10.0.0.0/16

Create VPC

IP pools [Info](#)

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ ap-south-1a (aps1-az1)

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and Internet gateway (IGW) routes configured to allow traffic.
subnet-0e4b21e444fd6c9 (Mumbai_VPC-subnet-public1-ap-south-1a)
10.0.0.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ ap-south-1b (aps1-az3)

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and Internet gateway (IGW) routes configured to allow traffic.
subnet-09bcd7034b05c112e (Mumbai_VPC-subnet-public2-ap-south-1b)
10.0.16.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ ap-south-1c (aps1-az2)

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently. Selected subnets must have Network ACL inbound and outbound ALLOW rules, and Internet gateway (IGW) routes configured to allow traffic.
subnet-030af62f99bad7ced (Mumbai_VPC-subnet-public3-ap-south-1c)
10.0.32.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

- Configure Listener and Routing
 - Choose listener port
 - Choose target group
 - Choose weight to move traffic

EC2 > Load balancers > Create Application Load Balancer

Listeners and routing [info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

Listener HTTP:80 [Remove](#)

Protocol
 HTTP

Port
 80
1-65535

Default action [info](#)
 The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups
 ☐ Redirect to URL
 ☐ Return fixed response

Forward to target group [info](#)
 Choose a target group and specify routing weight or [create target group](#).

Target group	Weight	Percent
Web-server-target-group Target type: Instance, IP4 Target stickiness: Off	HTTP 1 0-999	100%

[+ Add target group](#)
 You can add up to 4 more target groups.

Target group stickiness [info](#)
 Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional
 Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add listener tag](#)
 You can add up to 50 more tags.

- Verify ALB in AWS console

Load balancers (1/1) [What's new?](#) [Actions](#) [Create load balancer](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

☒	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS name
☒	web-server-ALB	Active	application	Internet-facing	IPv4	vpc-0d9bf3ad2f4293e02	3 Availability Zones	sg-0ef35bf8e55d77759	web-server-ALB

Load balancer: web-server-ALB

Details | Listeners and rules | Network mapping | **Resource map** | Security | Monitoring | Integrations | Attributes | Capacity | Tags

Resource map [info](#) [Give feedback](#)

View, explore, and troubleshoot your load balancer's architecture.

[Overview](#) | [Unhealthy target map](#) | ☒ Show resource details

web-server-ALB Last fetched seconds ago [Export](#)

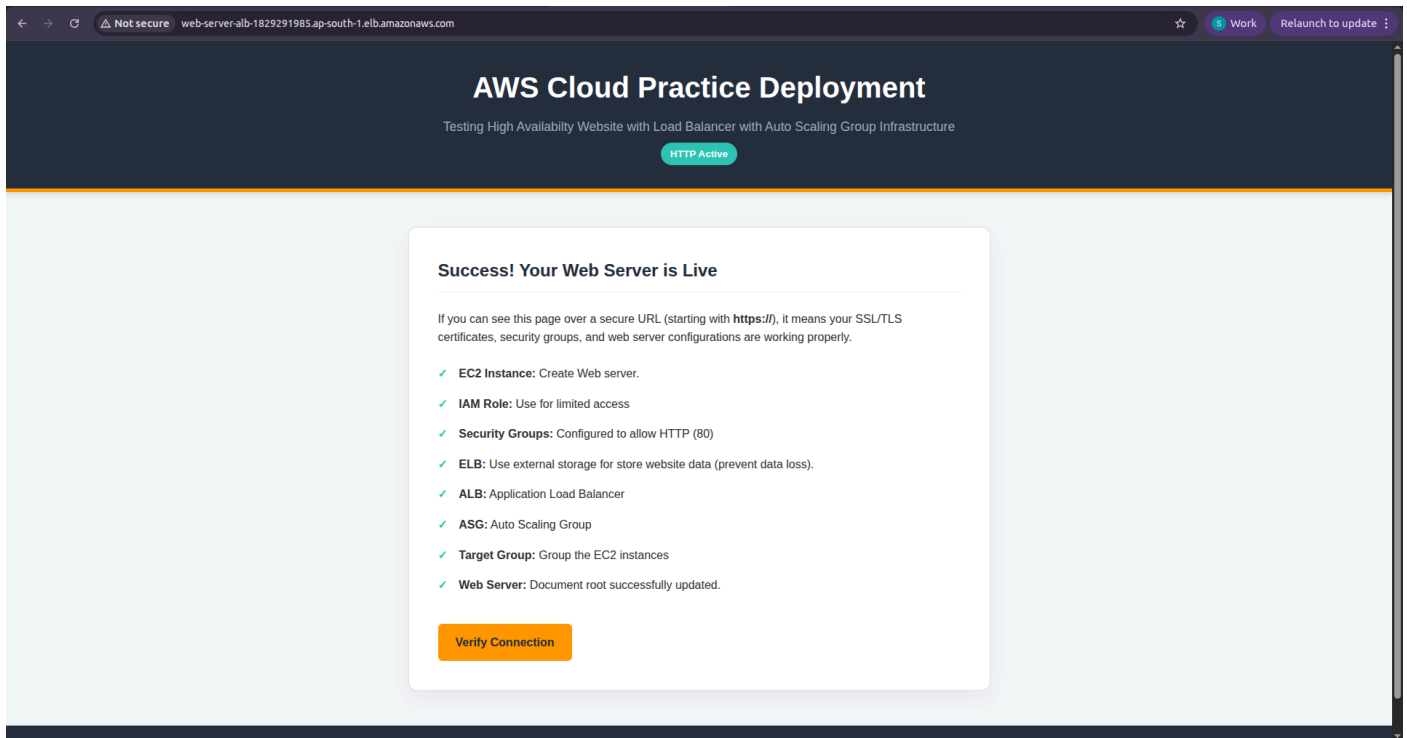
Listeners (1)
 HTTP:80

Rules (1)
 Priority default
 Forward to target group
 Conditions (if)
 If no other rule applies

Target groups (1) [info](#)
 Instance, HTTP
 Web-server-target-group
 2 targets
 2 0 0 0 0 0

Targets (2)
 i-0292df97a0dbfd401 Port 80
 Healthy
 i-09dfc79220ff24d6e Port 80
 Healthy

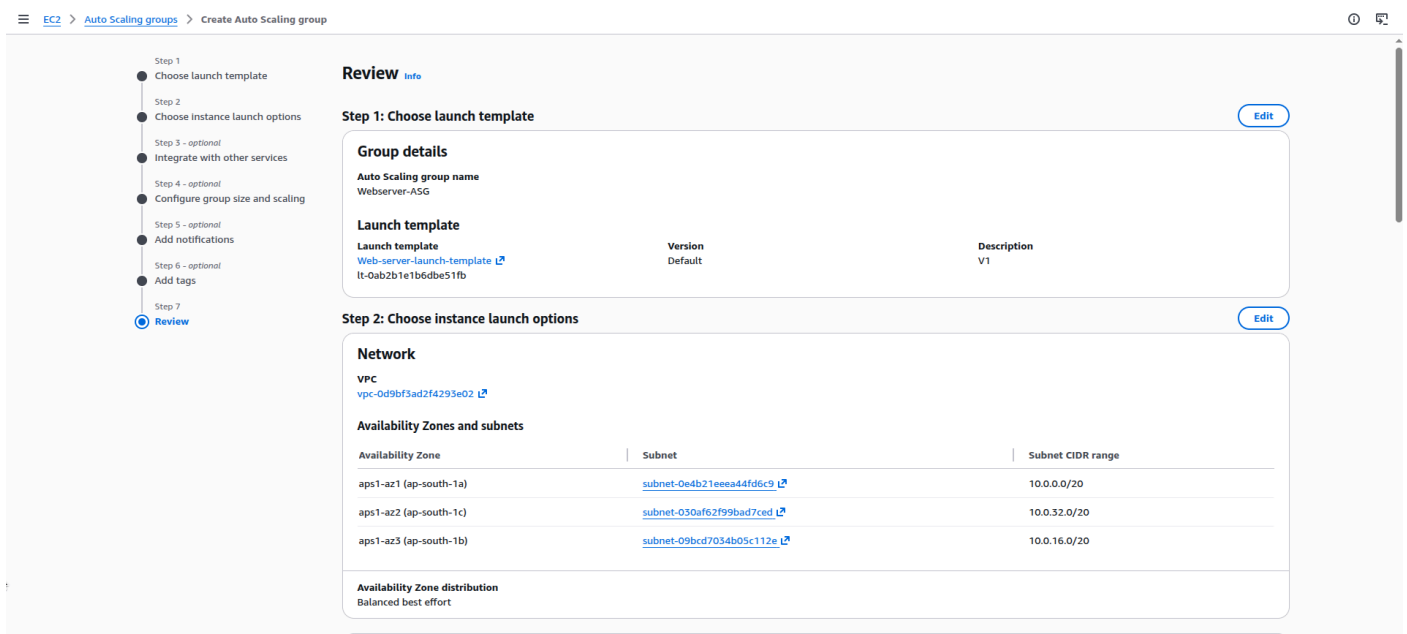
- Get confirmation website live or not (chose ALB DNS name and paste in chrome)



Phase 6

Create Auto Scaling Group

- Choose Launch template
- Choose Network setting and AZs
- Choose Load balancing
- Choose Load balancing target group
- Turn on ELB health check
- Choose health check period
- Choose group size (desire) and scaling size (min and max)



EC2 > Auto Scaling groups > Create Auto Scaling group

Step 3: Integrate with other services

Load balancing

Load balancer 1

Name	Web-server-ALB	Type	Application/HTTP	Target group	Web-server-target-group
------	----------------	------	------------------	--------------	-------------------------

VPC Lattice integration options

Application Recovery Controller (ARC) zonal shift

Health checks

Step 4: Configure group size and scaling policies

Group size

Desired capacity	2	Desired capacity type	Units (number of instances)
------------------	---	-----------------------	-----------------------------

Scaling

EC2 > Auto Scaling groups > Create Auto Scaling group

Scaling

Minimum desired capacity	2	Maximum desired capacity	4
Target tracking policy			

Instance maintenance policy

Replacement behavior	No policy	Min healthy percentage	-	Max healthy percentage	-
----------------------	-----------	------------------------	---	------------------------	---

Additional settings

Instance scale-in protection	Disabled	Monitoring	Disabled	Default instance warmup	Disabled
Placement group	-	Auto Scaling group deletion protection	None (default)		

Capacity Reservation preference

Preference	Default	Capacity Reservation IDs	-	Resource Groups	-
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Step 5: Add notifications

Notifications

Step 6: Add tags

- Verify ASG in AWS console

EC2 > Auto Scaling groups

Auto Scaling groups (1/1)

Launch configurationsLaunch templatesActionsCreate Auto Scaling group

Name	Status - new	Launch template/configuration	Instances	Instance health - new	Desired capacity	Min	Max	Availability Zones	Creation time
Webserver-ASG	Updating capacity	Web-server-launch-template	Version De 0	-	2	2	4	3 Availability Zones	Mon Jun 01 2026 ...

Auto Scaling group: Webserver-ASG

DetailsIntegrationsAutomatic scalingInstance managementInstance refreshActivityMonitoringTags

Webserver-ASG Capacity overview

arn:aws:autoscaling:ap-south-1:639713291467:autoScalingGroup:26293026-4de7-40f2-a036-3cbf742636c4:autoScalingGroupName/Webserver-ASG

Desired capacity	2	Scaling limits	2 - 4	Desired capacity type	Units (number of instances)	Status	Updating capacity
Date created	Mon Jun 01 2026 22:57:17 GMT+0530 (India Standard Time)						

Launch template

Launch template	AMI ID	Instance type	Owner
lt-0ab2b1e1b6d8e51fb Web-server-launch-template	ami-0127066faa399ce15	t2.nano	arn:aws:iam:639713291467:root
Version	Security groups	Security group IDs	Create time
Default	-	sg-02946fe197f932477	Mon Jun 01 2026 22:10:44 GMT+0530 (India Standard Time)

- Stop one EC2 to check fault tolerance and high availability

Successfully initiated stopping of i-09dfc79220ff24d6e

Instances (1/2) Info

Saved filter sets: Running

Instance state = running

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☐		i-0292df97a0dbfd401	Running	t2.nano	Initializing	ap-south-1b	ec2-3-108-235-147.ap-...	3.108.235.147	-	-
☒		i-09dfc79220ff24d6e	Stopping	t2.nano	Initializing	ap-south-1a	ec2-13-127-190-179.ap...	13.127.190.179	-	-

i-09dfc79220ff24d6e

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ **Instance summary** Info

Instance ID
i-09dfc79220ff24d6e

IPv6 address
-

Hostname type
IP name: ip-10-0-9-50.ap-south-1.compute.internal

Answer private resource DNS name
-

Public IPv4 address
13.127.190.179 | [open address](#)

Instance state
Stopping

Private IP DNS name (IPv4 only)
ip-10-0-9-50.ap-south-1.compute.internal

Instance type
t2.nano

Private IPv4 addresses
10.0.9.50

Public DNS
ec2-13-127-190-179.ap-south-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses
-

- In ALB go to unuse state

Load balancers (1/1) What's new?

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS name
☒	web-server-ALB	Active	application	Internet-facing	IPv4	vpc-0d9bf3ad2f4293e02	3 Availability Zones	sg-0ef35bf8e55d77759	web-server-ALB

Load balancer: web-server-ALB

Resource map Info

View, explore, and troubleshoot your load balancer's architecture.

Overview | Unhealthy target map | ☐ Show resource details

web-server-ALB Last fetched seconds ago

Listeners (1)

HTTP:80

Rules (1)

Priority default
Forward to target group

Conditions (if)
If no other rule applies

Target groups (1) Info

Instance, HTTP
Web-server-target-group

2 targets

1 0 1 0 0

Targets (2)

i-0292df97a0dbfd401 Port 80
Healthy

i-09dfc79220ff24d6e Port 80
Unused: Target is in the stopped state

- ASG Scale up new instance after stopping one

Load balancers (1/1) [What's new?](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS name
web-server-ALB	Active	application	Internet-facing	IPv4	vpc-0d9bf3ad2f4293e02	3 Availability Zones	sg-Def35bf8e55d77759	web-server-ALB

Load balancer: web-server-ALB

web-server-ALB

Last fetched seconds ago [Refresh](#) [Export](#)

Listeners (1)

HTTP:80

1 rule

Rules (1)

Priority default

Forward to target group

Conditions (If)

If no other rule applies

Target groups (1) [Info](#)

Instance, HTTP

Web-server-target-group

3 targets

2 Healthy 0 Degraded 0 Pending 0 Stopped 1 Inconsistent

Targets (3)

I-0292df97a0dbfd401 Port 80

Healthy

I-09dfc79220ff24d6e Port 80

Degrading: Target deregistration is in progress

I-0de966c28ef31101e Port 80

Healthy

- Terminate one instance

Successfully initiated termination (deletion) of I-0292df97a0dbfd401

Instances (1/2) [Info](#)

Notifications 0 0 2 0 0 0

Instance state [Instance state](#) [Actions](#) [Launch instances](#)

Running

Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
I-0292df97a0dbfd401	I-0292df97a0dbfd401	Shutting-down	t2.nano	2/2 checks passed	ap-south-1b	ec2-3-108-235-147.ap-...	3.108.235.147	-	-
I-0de966c28ef31101e	I-0de966c28ef31101e	Running	t2.nano	Initializing	ap-south-1a	ec2-15-206-167-101.ap...	15.206.167.101	-	-

i-0292df97a0dbfd401

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary [Info](#)

Instance ID

I-0292df97a0dbfd401

IPv6 address

-

Hostname type

IP name: ip-10-0-17-244.ap-south-1.compute.internal

Answer private resource DNS name

-

Public IPv4 address

3.108.235.147 [open address](#)

Instance state

Shutting-down

Private IP DNS name (IPv4 only)

ip-10-0-17-244.ap-south-1.compute.internal

Instance type

t2.nano

Private IPv4 addresses

10.0.17.244

Public DNS

ec2-3-108-235-147.ap-south-1.compute.amazonaws.com [open address](#)

Elastic IP addresses

-

- ALB show only one live instance

Load balancers (1/1) [What's new?](#)

Actions

Create load balancer

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

< 1 > ⚙

☒	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS name
☒	web-server-ALB	Active	application	Internet-facing	IPv4	ypc-0d9bf3ad2f4293e02	3 Availability Zones	sg-0ef35bf8e55d77759	web-server-ALB

Load balancer: web-server-ALB

Resource map [Info](#)

View, explore, and troubleshoot your load balancer's architecture.

Overview

Unhealthy target map

Show resource details

web-server-ALB

Last fetched seconds ago [Refresh](#) [Export](#)

Listeners (1)

HTTP:80

1 rule

Rules (1)

Priority default
Forward to target group

Conditions (if)
If no other rule applies

Target groups (1) [Info](#)

Instance, HTTP
Web-server-target-group

2 targets

1

0

0

0

1

Targets (2)

I-09dfc79220ff24d6e Port 80

Draining: Target deregistration is in progress

I-0de966c28ef31101e Port 80

Healthy

Load balancers (1/1) [What's new?](#)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

 Filter load balancers

<
1
>
⚙️

☒	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS name
☒	web-server-ALB	✔ Active	application	Internet-facing	IPv4	vpc-0d9bf3ad2f24293e02	3 Availability Zones	sg-Oef35bf8e55d77759.i	web-server-ALB

Load balancer: web-server-ALB

Last fetched seconds ago 🔄 Export

Listeners (1)

- HTTP:80 → 1 rule

Rules (1)

- Priority default
Forward to target group
- Conditions (If)
If no other rule applies

Target groups (1) Info

- Instance, HTTP Web-server-target-group 3 targets
- ✔ 2 🔍 0 🔄 0 🛑 0 ❌ 1

Targets (3)

- I-01a7925ead82aa402 Port 80 ✔ Healthy
- I-09dfc79220ff24d6e Port 80
- I-0de966c28ef31101e Port 80 ✔ Healthy
- 🔄 Draining: Target deregistration is in progress